

# Hongjie Zhu

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## EDUCATION

Northeastern University – University of Dundee | Biomedical Engineering

Aug 2023–Present

Weighted Average: 85/100 | Overall Grade: A | IELTS: 6.5 (minimum band score: 6.0)

## SELECTED PUBLICATIONS AND MANUSCRIPTS

### Published

- Sun H, **Zhu H**, et al. Real-time AI-guided ultrasound localization method for breast tumor rotational resection. *Frontiers in Oncology*, 2025, IF 3.4, JCR Q2.
- Shao X, **Zhu H**, et al. Real-time deep learning for tumor segmentation and tool tracking: development and validation of an AI navigation system in vacuum-assisted breast biopsy. *World Journal of Surgical Oncology*, 2025, IF 2.8, JCR Q1.
- **Zhu H**, et al. An AI-Assisted Clinical Decision Support System for Green Classification of Cystocele on Dynamic Transperineal Ultrasound. *Journal of Medical Systems*, 2026, IF 8.8, JCR Q1.
- **Zhu H**, et al. TriSCoV-Net: Cross-Scale Verified Virtual Immunomarker Proxy Generation from Diffusion Magnetic Resonance Imaging. *MICCAI 2026 Workshop CMMCA*, Oral Presentation.

### Under Review / Revision

- **Zhu H**, et al. Real-Time Tracking of Shrinking and Deforming Target in Ultrasound-Guided Vacuum-Assisted Breast Tumor Resection: A Lightweight Framework for Dual-Localization of Lesion and Instrument. *Biomedical Signal Processing and Control*, Minor Revision, IF 5.7, JCR Q2.
- **Zhu H**, et al. CEUS-Clinical Multimodal Model for Differentiating Hepatocellular Carcinoma from Non-HCC Malignancies in CEUS LI-RADS LR-M Lesions: Retrospective Development, Prospective Testing, and Reader Study. *Radiology: Imaging Cancer*, Major Revision, IF 7.2, JCR Q1.
- Zhou H, **Zhu H**, et al. A Non-invasive Multimodal Ultrasound–Clinical Model for Peri-transfer Pregnancy Risk Stratification after Single Embryo Transfer: A Temporal Validation Study. *Reproductive BioMedicine Online*, Under Review, IF 3.9, JCR Q1.

## RESEARCH EXPERIENCE

### Real-Time AI Navigation for Ultrasound-Guided Breast Tumor Resection

Mar 2024 – Present

Researcher

Supervisor: Prof. Hong Li

- Developed real-time dual-target perception and navigation for ultrasound-guided breast intervention, jointly localizing deforming lesions and fast-moving instrument tips.
- Curated a retrospective–prospective dataset of 167 patients and 10,668 intraoperative key frames; a lightweight detector achieved lesion and instrument-tip mAP50 of 0.861 and 0.796 in real time.
- Conducted an offline playback crossover reader study with 3 breast surgeons across 18 cases. AI overlays improved instrument-tip precision from 0.502-0.523 to 0.743-0.761 and reduced localization time by 58-63%.

### RCM-Constrained Bimanual Laparoscopic Robotic System

Jun 2026 – Present

Researcher

Supervisor: Prof. Fangxun Zhong

- Built a ROS 2 upper-layer control architecture for a CR5 6-DoF arm and self-built MPK 5-axis mechanism; transferred validated shared-control components to the physical platform for joint-space tracking and continuous motion.
- Integrated a monocular vision pipeline for target localization and robot control, validating autonomous ring localization, grasping, and transfer as a perception-to-action loop.
- Extending the platform toward bimanual coordination and RCM-constrained motion; partially reproduced selected components of prior surgical-instrument planning and control work.
- Built a ROS 2 simulation-and-control workspace for a six-DoF arm and gripper; configured RViz2, Gazebo Sim, and joint-trajectory control.
- Integrated an RGB-D camera for simulated sensor bridging and point-cloud coordinate correction, validating vision-based target grasping during robot motion.

- Robotic Perception and Geometry-Based Target Localization**

Sep 2023 – Sep 2024

Team Member
 Supervisor: Prof. Zhiguo Lu

  - Implemented an end-to-end pipeline for image acquisition, camera calibration, target detection, and 3D pose estimation.
  - Estimated target pose using SolvePnP and Kalman filtering to improve robustness against image noise and target motion.
- TriSCoV-Net: Virtual Immunomarker Proxy Generation from dMRI**

May 2025 – Present

Researcher
 Supervisor: Prof. Chao Li

  - Proposed a three-stage dMRI-to-IHC virtual immunomarker framework combining low-resolution synthesis, PLI-guided super-resolution, and cross-scale validation to address weak registration and scale mismatch.
  - Developed a shared dMRI encoder with slice-level global PLI priors, gated FiLM injection, and support-domain geometric-consistency constraints.
  - Processed Oxford Digital Brain Bank data and conducted subject-level LOSO evaluation with standardized ImageJ DAB quantification.
- LRM-Fusion: CEUS–Clinical Model for LI-RADS LR-M Differentiation**

Sep 2024 – Present

Researcher
 Supervisor: Prof. Hong Li

  - Designed an attention-based temporal imaging branch and clinical-information fusion strategy; achieved AUC 0.898 and accuracy 88.9% on prospective and external multicenter test cohorts.
  - Evaluated calibration, decision curves, interpretability, and a four-radiologist reader study; AI assistance improved all readers' accuracy and Fleiss'  $\kappa$  from 0.424 to 0.771.
- Automated Green Classification of Cystocele from Dynamic TPUS**

Sep 2025 – Present

Researcher
 Supervisor: Prof. Hong Li

  - Built an automated dynamic transperineal-ultrasound analysis system using key-anatomy localization, temporal continuity constraints, and attention-based temporal modeling for Green classification.
  - Established an 881-case retrospective–prospective cohort (688 development; 193 independent prospective test); achieved macro-AUC 0.939 and overall accuracy 90.2% prospectively.
- Peri-Transfer Multimodal Ultrasound–Clinical Pregnancy Risk Model**

Jul 2025 – Present

Researcher
 Supervisor: Prof. Hong Li

  - Contributed to a non-invasive IVF/ICSI pregnancy-outcome prediction framework integrating transvaginal B-mode ultrasound, Doppler perfusion, short cine dynamics, clinical variables, and embryo-development stage.
  - Designed and validated a gated multimodal fusion model; achieved AUC 0.771 and accuracy 0.747 on an independent temporal-validation cohort, supported by calibration, decision-curve, and SHAP analyses.

## HONORS AND AWARDS

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|----------------------------------------------------------------------------------------------------------------|------|
| • National First Prize, 11th National Biomedical Engineering Innovation Design Competition, 192 of 4,605 teams | 2026 |
| • National First Prize, 19th National College Student Software Innovation Competition, 15 of 2,392 teams       | 2026 |
| • Regional First Prize, 19th National College Student Software Innovation Competition (Northeast Region)       | 2026 |
| • National First Prize, 10th National Biomedical Engineering Innovation Design Competition, 145 of 4,200 teams | 2025 |
| • National First Prize, RoboMaster University Championship (National Finals) - 3rd of 32 national teams        | 2024 |
| • Regional First Prize, RoboMaster University Championship (Southern Region) - 2nd of 32 teams                 | 2024 |

## TECHNICAL SKILLS

- Mechanical Design & Prototyping: SolidWorks, CAD, Bambu Lab 3D printing.
- Robotics & Control: ROS 2, Gazebo, RViz2, CoppeliaSim, sim-to-real deployment, robot kinematics and control, RCM-constrained motion, bimanual coordination.
- Robot Vision: Camera calibration, hand-eye calibration, monocular pose estimation, OpenCV, RGB-D sensing and point-cloud processing for simulation.
- Medical AI: C++, Python, MATLAB, PyTorch, TensorRT, OpenVINO, real-time object detection, keypoint detection, segmentation, temporal modelling, multimodal fusion.